MILITARY SPECIFICATION

CABLE ASSEMBLY, SPECIAL PURPOSE, ELECTRICAL: W47U THROUGH W55U AND W65U AND W72U

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of cable assembly, entitled Cable Assembly, Special Purpose, Electrical: W47U, W48U, W49U, W50U, W51U, W52U, W53U, W54U, W55U, W65U and W72U. These cable assemblies are used with the Lance warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

8876233	- Cable Mandatory Process Information.
8876245	- Coaxial Cable and Connector Mandatory Process Information.
9236937	- Cable Control Unit to Station 100, Training W47U.

FSC 1135

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EL-9236937 - Equipment List.
9236938 - Cable Warhead, Training: W48U.
EL-9236938 - Equipment List.
9236939 - Cable, Fuze, Impact, Training: W49U.
9236940 - Cable, Fuze, Impact, Training: W50U.
EL-9236940 - Equipment List.
9236941 - Cable Fuze Command, Training: W51U.
EL-9236941 - Equipment List.
9236942 - Cable PAL, Internal, Training: W52U.
EL-9236942 - Equipment List.
9236943 - Cable, PAL, External, Training: W53U.
EL-9236943 - Equipment List.
9236981 - Cable, Impact Crystal, Training: W54U.
EL-9236981 - Equipment List.
9236982 - Cable, Impact Crystal, Training: W55U.
EL-9236982 - Equipment List.
9237235 - Cable Assembly, Option Selector Switch Training: W72U.
EL-9237235 - Equipment List.
9249024 - Cable, PAL, Special Purpose, Electrical, Type X, Training: W65U.
EL-9249024 - Equipment List.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The cable assemblies shall comply with all requirements specified on the drawing (dwg) 9236937, 9236938, 9236939, 9236940, 9236941, 9236942, 9236943, 9236981, 9236982, 9237235 and 9249024, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein after not less than 96 hours have elapsed upon completion of molding process.

3.2.1 Mating assembly. The cable assemblies shall be capable of being completely mated and unmated with their proper receptacles without the use of tools. This capability shall be verified as specified in 4.4.1.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

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3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 Circuit resistance. The circuit resistance between the pins of the cable assemblies listed in Tables I through VI shall be as specified when tested in accordance with 4.4.2.

Table I

W47U Cable, Control Unit to Station 100 (dwg 9236937)
Resistance 0.20 ohms (max)

From 1P2	To 1A1J1
B	L
H	M

Table II

W48U Cable, Warhead (dwg 9236938)
Resistance 0.20 ohms (max)

From 1P12	To 1P12
L	K

Table III

W51U Cable, Fuze, Command (dwg 9236941)
Resistance 0.20 ohms (max)

From 1P1	From 1P28	To 1P29
H		M
	M	E
B	E	

Table IV

W52U Cable, PAL (Internal) (dwg 9236942)
Resistance 0.20 ohms (max)

From A2P1	To A2J2	From A2P1	To A2J2
A	A	K	K
B	B	L	L
C	C	M	M

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E	E	P	P
F	F	R	R
G	G	S	S
H	H	U	U
D	D	Shell	Shell
N	N		

Table V

W53U Cable, PAL (External) (dwg 9236943)
Resistance 0.20 ohms (max)

From A2P2	To A2J3	From A2P2	To A2J3
A	A	L	L
B	B	M	M
C	C	N	N
D	D	P	P
E	E	R	R
F	F	S	S
G	G	U	U
H	H	Shell	Shell
K	K		

Table VI

W65U Cable, PAL (Assembly, Special Purpose, Electrical)
(Type X TNG) (dwg 9249024)
Resistance 0.20 ohms (max)

From A2P1	To A2J2	From A2P1	To A2J2
A	A	K	K
B	B	L	L
C	C	M	M
D	D	N	N
E	E	P	P
F	F	R	R
G	G	S	S
H	H	U	U

3.6 Insulation resistance. The insulation resistance between each circuit and all other circuits and connector shells connected together as one circuit and grounded, shall be not less than 50 megohms when 500 + 50 volts direct current (vdc) is applied. Testing shall be as specified in 4.4.3 with Tables VII through XII.

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Table VII

W47U Cable, Warhead (dwg 9236937)

1P2 - B, H

Table VIII

W48U Cable, Warhead (dwg 9236938)

1P12 - L

Table IX

W51U Cable Fuze, Command (dwg 9236941)

1P1 -- B, H
1P28 - M, E
1P29 - M, E

Table X

W52U Cable, PAL (Internal) (dwg 9236942)

A2P1 - A,B,C,D,E,F,G,H,K,L,M,N,P,R,S,U,Shell
A2P1 - Unused Terminal: J, T
A2J2 - Unused Terminal: J, T

Table XI

W53U Cable, PAL (dwg 9236943)

A2P2 - A,B,C,D,E,F,G,H,K,L,M,N,P,R,S,U,Shell
A2P2 - Unused Terminal: J, T
A2J3 - Unused Terminal: J, T

Table XII

W65U Cable, Assembly, Special Purpose, Electrical PAL (Type X) (dwg 9240924)

- A,B,C,D,E,F,G,H,J,K,L,M,N,P,R,S,T,U

3.7 Environmental testing. The cable assemblies shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The cable assemblies shall also comply with 3.5 and 3.6 before and after each environmental test. Testing shall be as specified in 4.4.4, 4.3.3.1 and 4.4.4. Tests shall be conducted in the order specified.

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3.7.1 Transportation vibration. The cable assembly shall be subjected in each of its three mutually perpendicular axes to the vibration levels for the lengths of times shown in Table XIII. After completion of the vibration tests, the cable assembly shall comply to the requirements of 3.5 and 3.6 and shall show no evidence of connector damage. The cable assembly shall be tested as specified in 4.4.4.1.

Table XIII

Frequency Hz*	Sweep Rate	Double Amplitude (da) or g Level
10-50-10	1 frequency sweep at a logarithmic rate between 10-500-10 in 15 minutes.	0.04 inch da 5.0 gravity units g's 10. g's
50-115-50		
115-500-115		

*cycles per second.

3.8 Handling.

3.8.1 Bend (W47U, W48U, W51U, W52U, W53U, W65U Cables). The cable assembly shall show no evidence of distortion, cracks, bend separation, or any other damage when subjected to flexing around mandrels specified in 4.4.5.1. Wrinkling of the cable jacket during flexure is acceptable. Cable shall bend freely around specified mandrel. Space between cable and mandrel shall not exceed 0.06 inches.

3.8.2 Twist-degrees (W47U, W48U, W51U, W52U, W53U, W65U Cables). The cable assembly shall show no evidence of cracks, mold bond separation, or any other damage when twisted as specified in 4.4.5.2. Cable assembly shall comply to requirements of 3.5 and 3.6 upon completion of tests. Distortion of cable jacket during twist is acceptable.

3.9 Soldering. All soldering shall comply with the requirement specified in MIL-STD-1460.

3.10 Workmanship.

3.10.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.10.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.10.3 Burrs. Parts shall be free from burrs.

3.10.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.10.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

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3.10.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 4 complete assemblies of W47U, W48U, W51U, W52U, W53U and W65U for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table XIV. Tests shall be conducted in order specified in Table XIV.

Table XIV

First article inspection or test

Inspection or Test	Dest	Non Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Circuit Resistance (see 3.5)		X
3. Insulation Resistance (see 3.6)		X
4. Transportation Vibration (see 3.7)		X
5. Handling (see 3.8)		X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

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4.3.1 Lot formation. The formation of an inspection lot and related functions shall be in accordance with Section 5, MIL-STD-105. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision.

4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table XV and Table XVI), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Table XV
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table XVI
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

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4.3.2.1.1 Cable assemblies - before molding (all applicable dwgs. - see 2.1).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Any component missing, damaged, loose or improper	Visual
102.	Improper connector	Visual
*103.	Improper soldering (see 3.9)	Visual
104.	Improper pre-pot cleaning	Visual
*105.	Improper attachment of shielding braid	Visual
106.	Presence of chips, foreign matter	Visual
107.	Evidence of poor workmanship (see 3.10)	Visual
108.	Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:	None defined.	

*Does not apply to W72U cable.

4.3.2.1.2 Cable assemblies - after molding (all applicable dwgs. - see 2.1).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Incorrect length	Gage
102.	Incorrect connector orientation	Visual
103.	Incorrect physical configuration	Visual
104.	Lead wire not completely covered by molded rubber	Visual
105.	Any component missing, damaged, loose or improper	Visual
106.	Marking incorrect, missing or illegible	Visual
107.	Poor bond between molded rubber and tubing	Visual
108.	Poor bond between molded rubber and connector shell	Visual
109.	Evidence of poor workmanship (see 3.10)	Visual
110.	Objective evidence relating to material missing or improper (see 3.1)	Visual

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Minor:
201. Serialization improper or missing (see 3.4).....Visual

4.3.2.1.3 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:
101. Preservation packing and packaging not in accordance with MIL-STD-1169(MU).....Visual
102. Marking not in accordance with MIL-STD-1169(MU).....Visual

Minor:
201. Evidence of poor workmanship (see 3.10).....Visual

4.3.3 Testing. All tests shall be performed. Tests need not be performed in the order specified below:

Test	Test Classification
a. Mating assembly	Major
b. Circuit resistance	Major
c. Insulation resistance	Major
d. Transportation vibration	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Mating assembly: W49U, W50U, W54U, W55U and W72U; circuit resistance and insulation resistance; W47U, W48U, W51U, W52U W53U and W65U. Each cable assembly shall be tested in accordance with 4.4.1, 4.4.2 and 4.4.3. If any item fails to comply with 3.2.1, 3.5 and 3.6, the item shall be rejected and removed from the lot.

4.3.3.2 Transportation vibration; W47U, W48U, W51U, W52U, W53U and W65U. A sample shall be selected at random in accordance with Table XVII: The sample shall be tested in accordance with 4.4.4.1. If any item fails to comply with 3.7.1, the lot shall be rejected.

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Table XVII

Sample sizes for non destructive testing

Lot Size	Sample Size	Lot Size	Sample Size
1-25	all	97-113	37
26-30	25	114-129	38
31-36	26	130-155	39
37-43	28	156-191	40
44-50	30	192-242	41
51-57	32	243-338	42
58-65	33	339-504	43
66-74	34	505-1120	44
75-85	35	1121 and over	45
86-96	36		

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-9236937, 9236938, 9236940, 9236941, 9236942, 9236943, 9236981, 9236982, 9237235 and 9249024). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Mating assembly. The cable assembly shall mate to connectors equivalent to the terminations they are to be used on so that visual observation of the locking pins, is confirmed, when proper coupling is accomplished.

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4.4.2 Circuit resistance. The resistance between the pins listed in 3.6 shall be measured with an approved electrical resistance tester. The resistance shall not exceed the values specified in 3.6. The current applied shall be 100 ± 1 milliamperes (ma).

4.4.3 Insulation resistance. Insulation resistance shall be measured by means of Government approved instrument capable of measuring megohms. Any circuit exhibiting an insulation resistance less than 50 megohms shall be accepted if the observed value increases to 50 megohms minimum after the specified voltage has been applied for 60 seconds. When measurements are made by observation of a meter needle, the range shall be selected so that the observed values lie within the central 3/4ths of that range. The current applied shall be limited to 4 ma maximum.

4.4.4 Environmental testings (see 3.7).

4.4.4.1 Transportation vibration. The cable assembly shall be rigidly mounted to an approved fixture capable of being readily positioned in each of the three mutually perpendicular axes. The cables shall be mounted to the approved fixture by attaching each cable connector to a mating connector mounted on the fixture. Spacing of the connectors on the fixture need not simulate final cable application position. The fixture shall then be mounted on a suitable vibration producing apparatus. The cable assembly shall then be vibrated at the frequencies, amplitudes and for the times specified in Table XIII.

4.4.5 Handling.

4.4.5.1 Bend; W47U, W48U, W51U, W52U, W53U, W65U Cables. Starting with jacket end and molding touching mandrel, bend the cable between 150° and 180° per portion, around mandrel. Remove this portion from the mandrel and repeat bends of 150° (min) with consecutive portions until entire unmolded portion of cable has been bent. Do not bend more than 180° any length of cable at one time.

Cable Assembly		Mandrel Dia (-0.06)
W51U	1P1 Branch	1.5
	1P28 Branch	1.5
	1P29 Branch	1.5
W48U, W52U, W53U		1.5
W47U		2.0

4.4.5.2 Twist (degrees) W47U, W48U, W51U, W52U, W53U, W65U Cables. Apply a torque to the cable using simulated connectors, or with the molded joint clamped so as to obtain a twist of 90° ± 5° per foot of cable length. This twist shall be applied first in a clockwise direction, and then in a counterclockwise direction after the cable has been returned to its initial at-rest position. All nonmolded portions of cable shall be tested.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The cable assemblies shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The cable assemblies shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Levels A and C. The cable assemblies shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspection which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1207A dated 10 October 1972.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1135-A208

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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